

CS & IT ENGINEERING



Theory of Computation

DFA

Lecture No.- 06



By- Venkat sir

Recap of Previous Lecture



Topic

DFA $\Rightarrow$ language $\Rightarrow$ Condition

Topic

length

exactly $n \rightarrow (n+2)$ states

at least $n \rightarrow (n+1)$ states

at most $n \rightarrow (n+2)$ states

divisible by $n \rightarrow n$ states



Topic : Deterministic Finite Automata

FORMAL DFA :

DFA is defined as

$$\text{DFA} = (Q, \Sigma, q_0, F, \delta)$$

Q : Finite set of states

Σ : Input alphabet

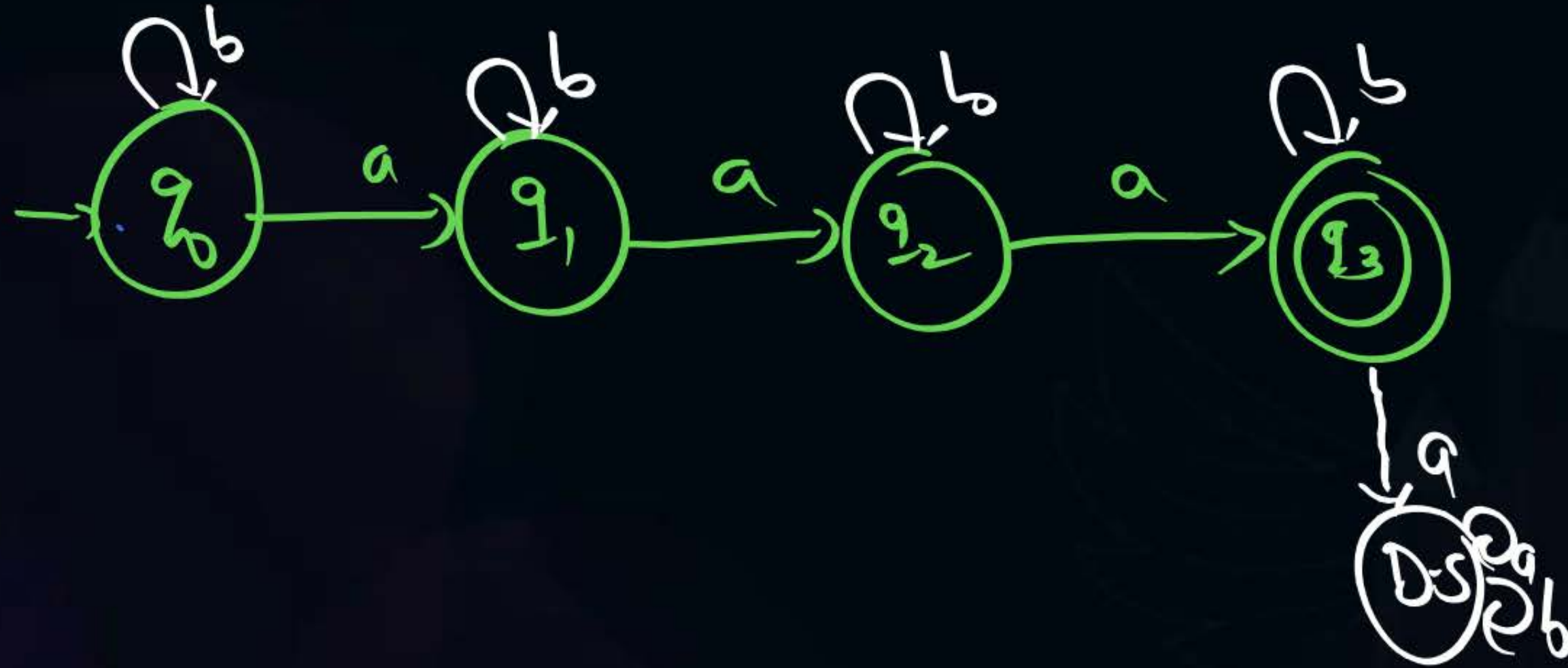
q_0 : Initial state

F : Set of final states

δ : Transition function $Q^* \Sigma \rightarrow Q$

(Q) How many states in min DFA over $\Sigma = \{a, b\}$ where
number of a's at most 4 } (6)

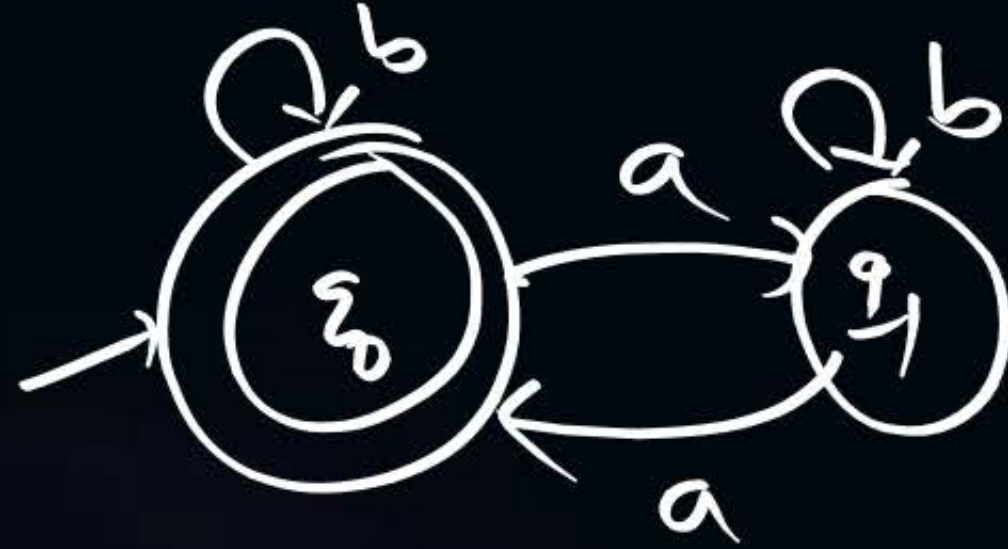
(n+2)



(Q)

$$\Sigma = \{a, b\}$$

a's div by 2 } min DFA states?



(Q)

a's div by 2

product DFA

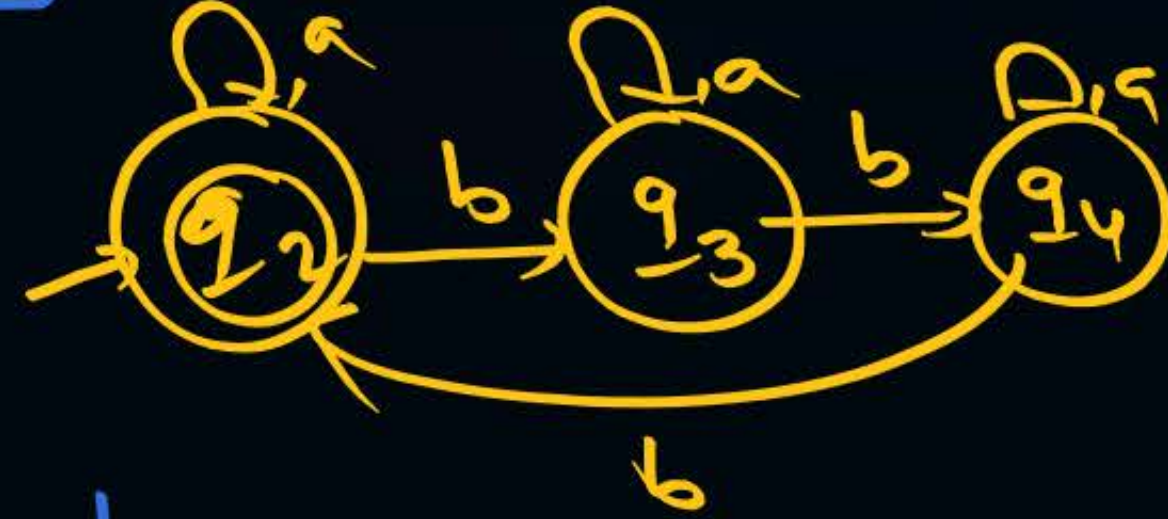
[00]
2 * 3

b's div by 3

minimal DFA ?



X



(a) 3

(b) 2

(c) 5

(d) 6

$Q = 6$

q_0

$F =$

δ

$\Sigma = \{a, b\}$



Topic : DFA

$$\Sigma = \{a, b\}$$

(h) # a's div by 6 [or] # b's div by 8 = (48) states

How many number of state are there with minimum DFA for the following state.

- (a) Number of a's ²divisible by 2 (and) ³number of b's not divisible by 3. = 6 states
- (b) Number of a's ²divisible by 2 (and) ⁴number of b's at least 3. = 8 States
- (c) Number of a's ³at least 2 (and) ⁴number of b's at least 3. = 12 states
- (d) Number of ³a's exactly 2 (and) ³number b's at least 2. = 10 states = $(9+1)$
- (e) Number of b's ⁴at most 3 (and) ⁴number b's exactly 3. = $16+1=17$ ✓
- (f) Number of a's ²not divisible by 2 (and) ⁴number of b's exactly 3. = $(8+1) = 9$
- (g) Number of a's div by 2 (or) Number of b's div by 3



Topic : DFA

NOTE:

Number of States of DFA on length conditions

- (i) Then in the given condition on length if one number divide other number then number of states of minimal DFA for "and" automata is LCM of given condition.
- (ii) Number of states of minimal DFA for "OR" automata is GCD of given condition.
- (iii) In the given length condition one number not divide other number then
→ If GCD of given condition is 1 then number of states of 'and' automata OR automata is multiplication of given condition.



Topic : DFA

- (iv) The given condition on length one number not divides other and GCD of given condition is not equal to 1 then number of states of 'and' automata, number of states of (OR) automata is LCM of given condition.

Find the number of stage of minimal DFA for the following matrix.

(Length of the string divisible by 3 or divisible by 6)

$$\text{GCD}(3, 6) = 3$$

Length of the string di is by 4 and di by 6

(v) $\text{LCM}(4, 6) = 12$

- (vi) Number of a's divisible by 4 AND number of b's divisible by 6.

$$4 \times 6 = 24$$



Topic : DFA

Find the number of stage of minimal DFA for the following matrix.

Length of the string divisible by 3 or divisible by 6

$$\text{GCD}(3, 6) = 3$$

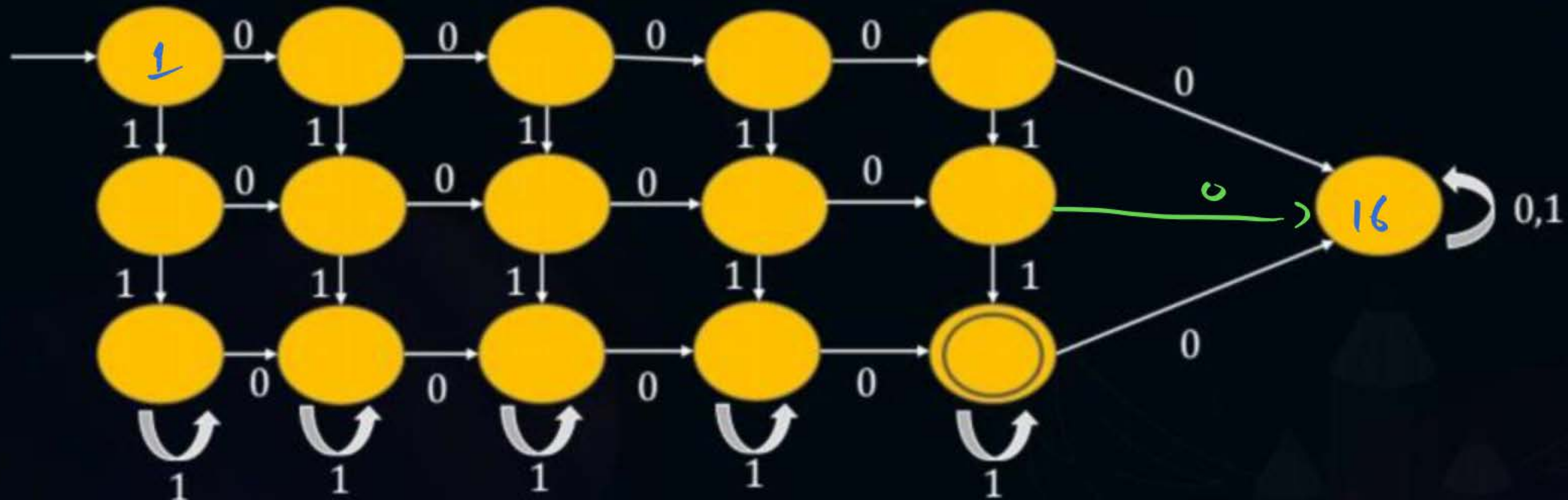
Length of the string di is by 4 and di by 6

$$\text{LCM}(4, 6) = 12$$

Number of a's divisible by 4 AND number of b's divisible by 6.

$$4 \times 6 = 24$$

#Q. Identify language accepted by following DFA



Set of all strings contains

- | | |
|--|--|
| A Length of the string atleast 6 | B # 0's exactly 4 and 1's atleast 3 |
| C # 0's atleast 4 and # 1's exactly 2 | D None |



Topic : DFA



$$\Sigma = \{a, b\}$$

$$LCM(2, 4) = 4$$

- ① Length of the string divisible by 2 and divisible by 4
- ② Length of string divisible by 2 (OR) divisible by 4. $\rightarrow \{0, 2, 4, 6, 8, 10, \dots\} = 2 \text{ states}$
 $GCD(2, 4) = 2$
- ③ Length of string divisible by 3 ^(and) divisible by 4 } # states $\Rightarrow LCM(3, 4) = 12$
 $GCD(3, 4) = 1$
- ④ Length of string divisible by 3 (OR) divisible by 4 } $\Rightarrow LCM(3, 4) = 12 \text{ states}$
- ⑤ Length of string divisible by 6 (OR) divisible by 8 $\Rightarrow LCM(6, 8) = 24 \text{ states}$
- ⑥ Number of a's divisible by 6 ⁶ (and) ⁸ number of b's divisible by 8. $\Rightarrow 48$
- ⑦ Length of the string div by 4 and div by 6 $\Rightarrow 12 \text{ } LCM(4, 6)$



2 mins Summary



Topic

One

DFA Constructions

Topic

Two

DFA $\rightarrow$ Language

Topic

Three

DFA minimization

Topic

Four

DFA Complement

Topic

Five



THANK - YOU